WORKSHOP ON AI APPLICATIONS IN EEG

31 Oct - 2 Nov, 2025
TLC, CENTRAL LIBRARY, IIT MADRAS, CHENNAI-600036

The human brain, one of the most complex biological systems, continuously generates electrical activity as we think, feel, move, or imagine. By capturing these signals using Electroencephalography (EEG), we open a gateway to interpreting brain states and building systems that interact directly with neural activity.

While EEG alone offers valuable insights into brain activity, its true potential is unlocked when combined with Artificial Intelligence (AI). Together, EEG and AI enable powerful applications in brain-computer interfaces, neurological diagnosis, cognitive monitoring, and beyond.

In this workshop, participants will explore how EEG signals are collected, processed, and analyzed using tools like MNE-Python. Working with EEG datasets, they will learn to build AI models for applications such as motor and visual imagery classification, seizure detection, and neurodegeneration analysis. The sessions include guided hands-on activities and expert talks offering a practical foundation in EEG-AI integration.

WORKSHOP HIGHLIGHTS

EEG data collection and preprocessing

Observe live demonstrations of EEG setup and signal acquisition
Learn how to preprocess raw neural signals using filtering, artifact removal, and segmentation techniques.

Machine Learning & Deep Learning for EEG Decoding

Explore how brain signals are decoded into meaningful outputs using ML/DL for tasks such as Motor & Visual Imagery decoding, Epileptic seizure detection, and analysis of neurodegenerative conditions such as AD and PD.

Live Software Sessions with MNE-Python

Gain practical experience in signal processing, Frequency analysis and ICA using MNE-Python.

Talks by Faculty & Neurotechnology Experts

Gain insights from leading researchers and industry professionals. Learn about cutting-edge EEG technologies, real-world applications, and current trends in BCI research from experienced faculty and neurotechnology experts.

SCHEDULE

DAY 1 : Foundations & EEG + AI Basics

(31st Oct 2025)

Time	Session
9.00-10.00	Inaugural Talk by Prof. Srinivasa Chakravarthy
10.00-11.00	**Introduction to EEG and system setup:** Overview of EEG: Principles and Applications Understanding the 10–20 montage, cap fitting and impedance check demonstration
11.00-12.00	**EEG data collection:** Resting state Recording: eyes open/closed
12.00-13.00	**EEG pre-processing:** Signal cleaning using Notch and Bandpass filter Artifact removal: Independent Component Analysis (ICA) using MNE-Python
13.00-14.00	Lunch Break
14.00-17.15	**Introduction to ML and DL for EEG:** Feature extraction: Power Spectral Density (PSD), Phase Lag Index (PLI) Models: XGBoost, CNN, LSTM, etc. Jupyter pipeline demo: walkthrough of an end-to-end classification task
17.15-17.30	Wrap-up and Q&A

Note:
Tea/refreshment breaks will be provided in both the morning and evening sessions. Exact timing may vary depending on the flow of the workshop each day.

DAY 2: Imagery & Clinical EEG Studies

(1st Nov 2025)

Time	Session
9.00-11.00	**Motor Imagery Theory and Lab:** Guided lab session using L/R hand imagination tasks Classification of Motor Imagery data offline
11.00-1.00	**Visual Imagery Theory and Lab:** Explore how imagined objects or scenes are reflected in EEG patterns Signal analysis and classification approaches
13.00-14.00	Lunch Break
14.00-15.30	**Epilepsy and Seizure Detection:** Introduction to BONN EEG Dataset Basics of spike detection and seizure event annotation
15.30-17.15	**Lab: Seizure Classification** Comparative modelling using Random forest and CNN architectures
17.15-17.30	Wrap-up and Q&A

DAY 3: Detecting EEG Markers and Neurodegenerative Disorders

(2nd Nov 2025)

Time	Session
9.00-11:00	Neurodegenerative Disorder Detection (Parkinson's vs Healthy); Extract Relevant EEG markers, Build and Evaluate ML Classifier
11:00-13.00	Neurodegenerative Disorder Detection (Alzheimer's vs Healthy); Extract Relevant EEG markers, Build and Evaluate ML Classifier
13.00-14.00	Lunch
14.00-17.15	**Talks by Faculty & Neurotechnology Experts** **Dr. Supratim Ray**, Prof., IISC Bangalore **Dr. Dipanjan Roy,** Assoc. Prof., IIT Jodhpur
17.15-17.30	Closing ceremony and Certificate distribution

Eligibility and Requirements

- Open to undergraduate and postgraduate students, researchers, and professionals.
- Basic familiarity with Python is helpful.
- Participants should bring their own laptop.

Fee details

UG Students --- 4000 INR
PG/PhD students --- 4500 INR
Professionals --- 5000 INR

Accommodation cost is not included. Relevant details are available on the website.

Registration Procedure

SOP submission by 21 September, 2025
Selection Results by 30 September, 2025
Registration payment by 7 October, 2025

How to apply?

Stage 1:

- Interested candidates must provide a Statement of Purpose (SOP) detailing their interest in the workshop.
- Based on the SOP, only **50** participants will be selected for the workshop.

SOP submission link

Stage 2:

- Selected participants will receive a confirmation email and payment link.
- Only confirmed participants are required to pay the registration fee.

What you will recieve

- Certificate of Participation
- Participant Handbook: Workshop on AI Applications In EEG
- Exclusive goodies

Our Contact

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